



Testing VPC Connectivity



ajv.singh3107@gmail.com

```
ec2-user@ip-10-0-1-113 ~$ curl nextwork.org
<a href="https://learn.nextwork.org/projects/aws-host-a-website-on-s3">Found</a>.
ec2-user@ip-10-0-1-113 ~$ curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>NextWork - Host a Website on Amazon S3</title>
  <meta content="Let's host your very own website on Amazon S3!" name="description" />
  <meta
    content="NextWork - Host a Website on Amazon S3"
    property="og:title"
  />
  <meta
    content="Let's host your very own website on Amazon S3!"
    property="og:description"
  />
  <meta
    content="https://cdn.prod.website-files.com/647460114a0547694b4afd16/650b837078acc7d663904eee_nextwork-opengraph-image.png"
    property="og:image"
  />
  <meta
    content="NextWork - Host a Website on Amazon S3"
    property="twitter:title"
  />
  <meta
    content="Let's host your very own website on Amazon S3!"
    property="twitter:description"
  />
  <meta
    content="https://cdn.prod.website-files.com/647460114a0547694b4afd16/650b837078acc7d663904eee_nextwork-opengraph-image.png"
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC lets you create a customizable, isolated network in AWS. It's useful for controlling network configurations, securing traffic with security groups and ACLs, and connecting securely to other networks, like on-premise systems.

How I used Amazon VPC in this project

I used Amazon VPC in today's project to set up a secure and isolated network environment for my application. I configured subnets, security groups, and route tables to control traffic and ensure connectivity between my resources and the internet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the complexity of configuring the security groups and route tables. Balancing security while ensuring proper connectivity for my resources required more adjustments and testing than I initially anticipated.

This project took me...

I completed this project in 1.5 hours, which involved configuring resources and troubleshooting various issues. The process required careful attention to detail to ensure optimal functionality and security within the Amazon VPC environment.



Connecting to an EC2 Instance

Connectivity means the ability of systems, devices, or networks to communicate with each other by sharing data. It refers to the links that enable digital interaction, such as through the internet, local networks, or wireless technologies like Wi-Fi

My first connectivity test was whether I could connect to Public Server.

```

Keyboard shortcut
To tab out of the terminal window and select the next button element, press the left and right Shift keys together.
Close permanently

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-109 ~]$ ifconfig
enx0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9001
    inet 10.0.1.109 netmask 255.255.255.0  broadcast 10.0.1.255
    inet6 fe80::26:96ff:fe4d:b59d prefixlen 64  scopeid 0x20<link>
    ether 02:26:96:4d:b5:9d  txqueuelen 1000  (Ethernet)
    RX packets 1284 bytes 177992 (173.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1646 bytes 170720 (166.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 12 bytes 1020 (1020.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1020 (1020.0 B)

i-Obbec9b3998025c7a (NextWork Public Server)
PublicIPs: 13.233.7.207  PrivateIPs: 10.0.1.109
```



EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is a browser-based SSH client provided by AWS. It allows secure, temporary SSH access to EC2 instances without needing a private key or a local SSH client, streamlining remote management.

My first attempt at getting direct access to my public server resulted in an error, because I had not correctly configured the security group to allow inbound SSH traffic from my IP address, blocking the connection request.

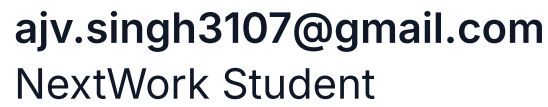
I fixed this error by updating the security group settings to allow inbound SSH traffic from my specific IP address, ensuring that the necessary port (typically 22) was open for secure connections to the public server.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-109 ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 10.0.1.109 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::165:56ff:fe4d:105a prefixlen 64 scopeid 0x20<link>
    ether 02:16:96:4d:10:5a txqueuelen 1000 (Ethernet)
    RX packets 1284 bytes 177992 (173.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 166 bytes 170720 (166.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 12 bytes 1020 (1020.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1020 (1020.0 B)
```

i-Obbec9b3998025C7a (NextWork Public Server)
PublicIP: 15.235.7.207 PrivateIP: 10.0.1.109



The ping command I ran was ping 10.0.2.168

```
[ec2-user@ip-10-0-1-109 ~]$
[ec2-user@ip-10-0-1-109 ~]$
[ec2-user@ip-10-0-1-109 ~]$
[ec2-user@ip-10-0-1-109 ~]$
[ec2-user@ip-10-0-1-109 ~]$ ping 10.0.2.253
PING 10.0.2.253 (10.0.2.253) 56(84) bytes of data.
```



Troubleshooting Connectivity

I troubleshooted this by allowing ICMP traffic in the private subnet ACL, which resolved the issue by enabling the necessary network communication for ping requests to reach the target device.

The screenshot shows an AWS terminal window with a dark background. At the top, there's a header bar with 'AWS', 'Services', a search bar, and '[Alt+S]'. Below this, a blue banner displays a 'Keyboard shortcut' message: 'To tab out of the terminal window and select the next button element, press the left and right Shift keys together.' with a 'Close permanently' button. The main terminal area shows the output of a series of ping commands. Each line starts with '64 bytes from 10.0.2.168: icmp_seq=' followed by a sequence number from 93 to 118, 'ttl=127', and 'time=' followed by a response time in milliseconds. All responses are successful, indicating connectivity.

```
64 bytes from 10.0.2.168: icmp_seq=93 ttl=127 time=1.27 ms
64 bytes from 10.0.2.168: icmp_seq=94 ttl=127 time=1.02 ms
64 bytes from 10.0.2.168: icmp_seq=95 ttl=127 time=0.693 ms
64 bytes from 10.0.2.168: icmp_seq=96 ttl=127 time=1.15 ms
64 bytes from 10.0.2.168: icmp_seq=97 ttl=127 time=1.19 ms
64 bytes from 10.0.2.168: icmp_seq=98 ttl=127 time=1.10 ms
64 bytes from 10.0.2.168: icmp_seq=99 ttl=127 time=1.45 ms
64 bytes from 10.0.2.168: icmp_seq=100 ttl=127 time=1.08 ms
64 bytes from 10.0.2.168: icmp_seq=101 ttl=127 time=1.36 ms
64 bytes from 10.0.2.168: icmp_seq=102 ttl=127 time=1.31 ms
64 bytes from 10.0.2.168: icmp_seq=103 ttl=127 time=1.08 ms
64 bytes from 10.0.2.168: icmp_seq=104 ttl=127 time=1.08 ms
64 bytes from 10.0.2.168: icmp_seq=105 ttl=127 time=0.962 ms
64 bytes from 10.0.2.168: icmp_seq=106 ttl=127 time=1.05 ms
64 bytes from 10.0.2.168: icmp_seq=107 ttl=127 time=1.43 ms
64 bytes from 10.0.2.168: icmp_seq=108 ttl=127 time=1.18 ms
64 bytes from 10.0.2.168: icmp_seq=109 ttl=127 time=1.13 ms
64 bytes from 10.0.2.168: icmp_seq=110 ttl=127 time=1.39 ms
64 bytes from 10.0.2.168: icmp_seq=111 ttl=127 time=1.75 ms
64 bytes from 10.0.2.168: icmp_seq=112 ttl=127 time=1.03 ms
64 bytes from 10.0.2.168: icmp_seq=113 ttl=127 time=1.36 ms
64 bytes from 10.0.2.168: icmp_seq=114 ttl=127 time=0.840 ms
64 bytes from 10.0.2.168: icmp_seq=115 ttl=127 time=1.02 ms
64 bytes from 10.0.2.168: icmp_seq=116 ttl=127 time=1.05 ms
64 bytes from 10.0.2.168: icmp_seq=117 ttl=127 time=1.33 ms
64 bytes from 10.0.2.168: icmp_seq=118 ttl=127 time=1.02 ms
```



Connectivity to the Internet

Curl is a command-line tool used to transfer data to or from a server using various protocols such as HTTP, HTTPS, FTP, and more. It's commonly used to test API endpoints, download files, and interact with web services.

I used curl to test the connectivity between my system and a web server, ensuring that the server was responding correctly to HTTP requests and verifying that the API endpoints were functional.

Ping vs Curl

Ping and curl are different because ping checks basic network connectivity by sending ICMP Echo Requests, while curl tests higher-level connectivity by interacting with web servers using protocols like HTTP/HTTPS to retrieve or send data.



Connectivity to the Internet

I ran the curl command `curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3` which returned the HTML content of the webpage, confirming successful connectivity and access to the website.

```
[ec2-user@ip-10-0-1-113 ~]$ curl nextwork.org
<a href="https://learn.nextwork.org/projects/aws-host-a-website-on-s3">Found</a>.

[ec2-user@ip-10-0-1-113 ~]$ curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3
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  <head>
    <meta charset="UTF-8" />
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    <meta content="Let's host your very own website on Amazon S3!" name="description" />
    <meta
      content="NextWork - Host a Website on Amazon S3"
      property="og:title"
    />
    <meta
      content="Let's host your very own website on Amazon S3!"
      property="og:description"
    />
    <meta
      content="https://cdn.prod.website-files.com/647460114a0547694b4afd16/650b837078ace7d663904eee_nextwork-opengraph-image.png"
      property="og:image"
    />
    <meta
      content="NextWork - Host a Website on Amazon S3"
      property="twitter:title"
    />
    <meta
      content="Let's host your very own website on Amazon S3!"
      property="twitter:description"
    />
    <meta
      content="https://cdn.prod.website-files.com/647460114a0547694b4afd16/650b837078ace7d663904eee_nextwork-opengraph-image.png"
```



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